Chapter 3

Homework 4 due: Thursday 27th March, 2014

All non-starred exercises from sections 3.1 and 3.2 (3.1-3.11).

Exercise 3.1. Prove that If $a|b$ and $b|c$ then $a|c$.

Proof. $a|b \Rightarrow b = na$ for some $n \in \mathbb{Z}$, and $b|c \Rightarrow mb = m\,(na) = c$ for some $m \in \mathbb{Z}$. Thus $a|c$ □

Exercise 3.2. If $a|b$ and $a|c$ then $a|b \pm c$.

Proof. Let $a|b$ and $a|c$. So then $b = an$ and $c = am$ for some $n, m \in \mathbb{Z}$. Thus $b \pm c = an \pm am = a\,(n \pm m)$ which implies that $a|b \pm c$. □

Exercise 3.3. ★ Prove that there are infinitely many prime numbers.

Exercise 3.4. Determine the following:

1. $\gcd(2^4 3^2 5 \cdot 7^2, 2 \cdot 3^3 7 \cdot 11)$

$$= 2 \cdot 3^2 \cdot 7$$

2. The lowest common multiple: $\operatorname{lcm}(2^4 3^2 5 \cdot 7^2, 2 \cdot 3^3 7 \cdot 11)$.

$$= 2^4 \cdot 3^3 \cdot 5 \cdot 7^2 \cdot 11$$

Exercise 3.5. ★ Suppose that $a \neq 0$, $b \neq 0$, and $\operatorname{lcm}(a, b) = m$. Prove that if n is a common multiple of a and b then $m|n$.

Exercise 3.6. ★ Let $a \neq 0$, $b \neq 0$, and $d = \gcd(a, b)$. Prove that if d' is any common divisor of a and b then $d'|d$.

Exercise 3.7. Find integers s and t so that $1 = 7 \cdot s + 11 \cdot t$. Show that s and t are not unique.

Proof. Consider $s = -3$ and $t = 2$. These can be found as follows:

$11 = 7 \cdot 1 + 4$
$7 = 4 \cdot 1 + 3$
$4 = 3 \cdot 1 + 1$
$3 = 1 \cdot 3 + 0$ so $1 = \gcd(11, 7)$

$1 = 3 \cdot (-1) + 4$
$1 = [7 - 4 \cdot 1]\,(-1) + 4 = 7(-1) + 4(2)$
$1 = 7(-1) + [11 - 7]\,(2)$
$1 = 7(-3) + 11(2)$

To see that s and t are unique, consider $s = 8$ and $t = -5$. Note that $as + bt = as + [abk - abk] + bt = a(s + bk) + b(t - ak)$ for any $k \in \mathbb{Z}$. Using the answer above, this gives us an infinite class of solutions. For $k = 1$, $7(-3 + 11) + 11(2 - 7) = 7(8) + 11(-5) = 1$. □

Equations of the form $ax + by = d$ are called Diophantine equations (after Diophantus of Alexandria 200-284 AD). Diophantine equations have a solution if and only if the $\gcd(a, b)|d$, and if a solution exists then there are infinitely many of them.

Exercise 3.8. Let $d = \gcd(a, b)$. If $a = da'$ and $b = db'$, show that $\gcd(a', b') = 1$.

Proof. By the GCD is a linear combination theorem, $d = ax + by$ for some $x, y \in \mathbb{Z}$. This implies that $d = da'x + db'y$, and so dividing by d gives $1 = a'x + b'y$. Thus $1 = \gcd(a', b')$ $\square$

Exercise 3.9. ★ If $c^2 = ab$ and $\gcd(a, b) = 1$, prove that a and b are perfect squares (Hint: By the theorem above, a and b can be written as the product of primes. They are perfect squares if and only if each prime in this prime factorization has an even power).

Exercise 3.10. Determine $51 \mod 13$, $342 \mod 85$, and $(82 \cdot 73) \mod 7$.

$51 = 3 \cdot 13 + 12$ so $12 \equiv 51 \mod 13$

$342 = 4 \cdot 85 + 2$ so $2 \equiv 342 \mod 85$

$82 \cdot 73 = 5986 = 855 \cdot 7 + 1$ so $1 \equiv (82 \cdot 73) \mod 7$

Exercise 3.11. Let n be a positive integer greater than 1 and let a and b be positive integers. Prove that $a \mod n \equiv b \mod n$ if and only if $a \equiv b \mod n$.

To prove a "**A** if and only if **B**" statement, you must show that **A** implies **B** and **B** implies **A** ("if and only if" is also written "iff." and $\iff$).

We discussed two equivalent ways to define $a = b \bmod n$:
$a = b \bmod n \iff n|b - a$, and
$a = b \bmod n \iff a = qn + b$ for some $q \in \mathbb{Z}$.

Proof. ($a \bmod n = b \bmod n \Rightarrow a = b \bmod n$) If $a \bmod n = b \bmod n$ then $n|\,(a - b \bmod n)$ and $n|\,(b - a \bmod n)$. So then $n|\,([a - b \bmod n] - [b - a \bmod n])$, and since $-b \bmod n + a \bmod n = 0$ the two middle terms cancel out implying that $n|\,(a - b)$. Thus $a = b \bmod n$.

($a = b \bmod n \Rightarrow a \bmod n = b \bmod n$) Suppose that $a = b \bmod n$ so then $n|b - a$. Since $n|a - a$, $a = a \bmod n$. It follows that $n|b - a \Rightarrow n|b - a \bmod n \Rightarrow a \bmod n = b \bmod n$. $\square$

(Alternatively)

Proof. ($a \bmod n = b \bmod n \Rightarrow a = b \bmod n$) If $a \bmod n = b \bmod n$ then $a = nq_1 + r$ and $b = nq_2 + r \Rightarrow a - b = n(q_1 - q_2) \Rightarrow n|a - b \Rightarrow a = b \bmod n$.

($a = b \bmod n \Rightarrow a \bmod n = b \bmod n$) Suppose that $a = b \bmod n$ then $a = q_1 n + r_1$ and $b = q_2 n + r_2$. Since $a = b \bmod n \Rightarrow n|a - b \Rightarrow n|n\,(q_1 - q_2) + (r_1 + r_2)$. Thus n divides $n\,(q_1 - q_2)$and also divides $r_1 + r_2$. Since $0 \leq r_1, r_2 < n$ it follows that $r_1 - r_2 = 0 \Rightarrow r_1 = r_2$. Thus $a \bmod n = b \bmod n$.

$\square$